Hints: Series and infinite products – 1. Convergence: the standard scale and comparison

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Condensation: cut $[1,\infty)$ into the blocks $2^k \leqslant n < 2^{k+1}$. A block has 2^k terms, each between $a_{2^{k+1}}$ and a_{2^k}, so

$$\tfrac{1}{2}\sum_{k\geqslant 0} 2^{k+1}a_{2^{k+1}} \leqslant \sum_{n\geqslant 1} a_n \leqslant \sum_{k\geqslant 0} 2^k a_{2^k}.$$

Integral test: $f(k+1) \leqslant \int_k^{k+1} f \leqslant f(k)$. The sequence $D_n = \sum_{k\leqslant n} f(k) - \int_1^n f$ is non-increasing ($D_{n+1} - D_n = f(n+1) - \int_n^{n+1} f \leqslant 0$) and $D_n \geqslant f(n) \geqslant 0$, $D_1 = f(1)$.

Hint 2. For $a_n = \frac{1}{n\ln^p n}$ condensation gives $2^k a_{2^k} = \frac{1}{(k\ln 2)^p}$, so the series converges exactly for $p > 1$. For $\frac{1}{n\ln n \ln^p \ln n}$ condensation gives terms $\asymp \frac{1}{k\ln^p k}$, which is the previous series; again $p > 1$. (The integral test does the same with the substitutions $t = \ln x$, $t = \ln\ln x$.)

Hint 3. No slowest convergent series. Let $r_n = \sum_{k\geqslant n} a_k$ be the tails, $r_n \downarrow 0$, and put $b_n = a_n/\sqrt{r_n}$. Then $b_n/a_n \to \infty$, and

$$b_n = \frac{r_n - r_{n+1}}{\sqrt{r_n}} \leqslant 2\bigl(\sqrt{r_n} - \sqrt{r_{n+1}}\bigr),$$

which telescopes, so $\sum b_n \leqslant 2\sqrt{r_1}$.

Hint 4. No slowest divergent series. Let $S_n = \sum_{k\leqslant n} a_k \to \infty$ and $b_n = a_n/\sqrt{S_n}$, so $b_n/a_n \to 0$. Now $\sqrt{S_n} - \sqrt{S_{n-1}} = \frac{a_n}{\sqrt{S_n}+\sqrt{S_{n-1}}} \leqslant b_n$, and the left-hand sides add up to $\sqrt{S_n} - \sqrt{S_1} \to \infty$.

Problem 2

Hint 1. One direction is a single choice of c_n: which c_n makes $\frac{a_n}{c_n}$ and $\frac{c_n}{b_n}$ equal?

Hint 2. $c_n = \sqrt{a_n b_n}$ makes both equal to $\sqrt{a_n/b_n}$, so 2) implies 1).

Hint 3. For 1) $\Rightarrow$ 2) write $\sqrt{\frac{a_n}{b_n}} = \sqrt{\frac{a_n}{c_n}\cdot\frac{c_n}{b_n}} \leqslant \frac{1}{2}\Bigl(\frac{a_n}{c_n} + \frac{c_n}{b_n}\Bigr)$ by AM–GM.

Problem 3

Hint 1. The series converges. It is enough to show $a_n \leqslant \frac{2}{3}n$ for all composite $n \geqslant 9$; the finitely many terms $n = 4, 6, 8$ do not matter, and then the terms are at most $(2/3)^n$.

Hint 2. If $n = ab$ with $1 < a < b$, then a and b are distinct factors of $b!$, so $n \mid b!$ and $a_n \leqslant b \leqslant n/2$. Every composite n has such a factorisation unless $n = p^2$ with p prime (take $a =$ the least prime divisor).

Hint 3. For $n = p^2$ the factorial $(2p)!$ contains the factors p and $2p$, so $a_n \leqslant 2p = 2\sqrt{n}$, and $2\sqrt{n} \leqslant \frac{2}{3}n$ for $n \geqslant 9$.

Problem 4

Hint 1. Compare the term with $\min(a_n, \frac{1}{n})$: if $na_n \leqslant 1$, then $\frac{a_n}{1+na_n} \geqslant \frac{a_n}{2}$; if $na_n \geqslant 1$, then $\frac{a_n}{1+na_n} \geqslant \frac{a_n}{2na_n} = \frac{1}{2n}$. So it suffices to show that $\sum \min(a_n, \frac{1}{n})$ diverges.

Hint 2. $b_n = \min(a_n, \frac{1}{n})$ is non-increasing, so condensation applies: $\sum b_n$ converges iff $\sum_k \min(2^k a_{2^k}, 1)$ does.

Hint 3. If the condensed series converged, its terms would tend to 0, so from some point on they equal $2^k a_{2^k}$, and $\sum 2^k a_{2^k}$ would converge. By condensation again $\sum a_n$ would converge, a contradiction.

Problem 5

Hint 1. First check by induction that $0 < x_{n+1} < x_n < 1$: $x_{n+1} = x_n(1 - \frac{x_n}{\sqrt{n}})$ and $x_n < 1 \leqslant \sqrt{n}$. Now find the asymptotics of x_n. For $x' = -x^2/\sqrt{t}$ one looks at $1/x$.

Hint 2. $\frac{1}{x_{n+1}} - \frac{1}{x_n} = \frac{x_n - x_{n+1}}{x_n x_{n+1}} = \frac{1}{\sqrt{n} - x_n}$. Since $0 < x_n < 1$, this is $\frac{1}{\sqrt{n}}\big(1 + O(n^{-1/2})\big)$.

Hint 3. By Stolz, $\frac{1/x_n}{\sqrt{n}}$ has the limit $\lim \frac{1/(\sqrt{n} - x_n)}{\sqrt{n+1} - \sqrt{n}} = 2$. So $x_n \sim \frac{1}{2\sqrt{n}}$, and $\sum x_n^\alpha$ converges exactly for $\alpha > 2$.